

Design and Implementation of a Wazuh-Based SOC Home Lab for Attack Detection and Log Analysis

A Comprehensive Technical Investigation into Endpoint Telemetry, eBPF Kernel Monitoring, Multi-Vector Cyber Attack Detection, and SIEM Alert Correlation

Author / Researcher:	Natto Muni Chakma
Degree Program:	B.Tech in Computer Science and Engineering
Institution:	Andhra University College of Engineering
Domain Specialization:	Cybersecurity, SIEM, SOC Operations, Threat Detection
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EXECUTIVE ABSTRACT: This research report details the architectural design, deployment, and validation of an enterprise-grade Security Operations Center (SOC) home laboratory. Built entirely on open-source and standard enterprise security platforms—primarily the Wazuh SIEM ecosystem integrated with Microsoft Sysmon on Windows and eBPF-based Sysmon for Linux—this project proves end-to-end detection engineering. Multiple cyberattack scenarios (network reconnaissance, SSH/RDP brute-force, unauthorized PowerShell execution, and persistence techniques) were simulated via Kali Linux and detected in real-time, resulting in over 11,000+ telemetry events and actionable alerts mapped against the MITRE ATT&CK framework.

1. Introduction & Project Scope

Modern enterprise cyber defense relies heavily on Security Information and Event Management (SIEM) solutions coupled with Security Operations Center (SOC) workflows. As cyber threats increase in stealth and sophistication, traditional signature-based antiviruses fail to provide sufficient visibility into advanced threat actor behavior, fileless attacks, and lateral movement.

The primary aim of this capstone project is to engineer an enterprise-grade, cost-effective SOC Home Lab. By deploying Wazuh as the centralized SIEM/XDR platform alongside deep kernel telemetry collectors (Microsoft Sysmon and Sysmon for Linux), this lab bridges theoretical cybersecurity knowledge with practical offensive and defensive capabilities.

Key Objectives

- **Architecture Design:** Establish an isolated, multi-node virtual network using Oracle VirtualBox NAT Networking.
- **SIEM & XDR Deployment:** Install, configure, and harden Ubuntu Server 24.04 hosting Wazuh Manager, Wazuh Indexer (OpenSearch), and Wazuh Dashboard.
- **Endpoint Telemetry Integration:** Deploy Wazuh Agents with Microsoft Sysmon on Windows 11 and eBPF-driven Sysmon for Linux.
- **Offensive Attack Simulation:** Execute realistic multi-stage cyber attacks from a Kali Linux adversary node.
- **Detection Engineering:** Author custom decoders and XML rules, validating detection triggers against MITRE ATT&CK; tactics.

2. Lab Architecture & Node Specifications

The lab is structured inside an isolated VirtualBox NAT Network (10.0.2.0/24 subnet), preventing accidental external network leakage while enabling unrestricted inter-node communication for realistic attack traffic and encrypted log forwarding.

Node / Hostname	Operating System	Role in SOC Lab	Assigned IP	Resources
Wazuh Server	Ubuntu Server 24.04 LTS	SIEM Manager, Indexer & Dashboard	10.0.2.15	4 GB RAM, 2 vCPU, 50 GB Disk
Victim Endpoint	Windows 11 Pro (64-bit)	Primary Monitored Workstation	10.0.2.20 / DHCP	4 GB RAM, 2 vCPU, 60 GB Disk
Attacker Node	Kali Linux (Rolling)	Offensive Attack & Recon Simulation	10.0.2.30 / DHCP	2 GB RAM, 2 vCPU, 30 GB Disk
Host Machine	Windows 11 Pro (Physical)	Hypervisor Host (Oracle VirtualBox)	Host Gateway	16 GB RAM, Intel Core

3. Wazuh SIEM Platform Deployment & Services

Wazuh operates as an all-in-one centralized architecture comprising three critical microservices:

1. **Wazuh Manager:** Receives encrypted telemetry from agent daemons on TCP port 1514, handles agent registration on port 1515, executes real-time decoder matching, and processes the correlation rule engine.
2. **Wazuh Indexer:** A high-performance, distributed OpenSearch engine providing scalable document indexing, document storage, and multi-field querying across millions of security events.

3. **Wazuh Dashboard:** A rich React-based web user interface operating over HTTPS (port 443) that provides interactive visual analytics, threat hunting panels, and compliance reporting.

Service Verification & Health Checks

Following deployment, all system daemons were verified using standard systemd utilities, confirming zero runtime errors across `wazuh-manager`, `wazuh-indexer`, and `wazuh-dashboard`.

4. Endpoint Telemetry & Sysmon Integration (Windows & Linux)

Default operating system event logging often fails to record critical process lineage, command-line arguments, and socket-to-process bindings. To bridge this visibility gap, Microsoft Sysmon was deployed across both Windows and Linux endpoints.

Windows 11 Sysmon Architecture

Sysmon operates as a Windows system service and device driver that remains resident across system boots. The Wazuh Agent configuration (`ossec.conf`) was updated to subscribe directly to the `Microsoft-Windows-Sysmon/Operational` event channel using the native Windows Event Log API.

Sysmon for Linux (eBPF) Architecture

On Linux endpoints, `sysmonforlinux` utilizes extended Berkeley Packet Filter (eBPF) programs injected into kernel tracepoints. This enables non-intrusive, microsecond-latency capture of process lifecycle events and socket connections without requiring kernel modifications. Sysmon events stream directly into `/var/log/syslog` and are ingested by the Linux Wazuh Agent.

Event ID	Event Name	Platform	Operational SOC Value & Detection Coverage
1	Process Creation	Win & Linux	Captures full CLI arguments, ParentProcessId, ProcessGuid, hashes, user context.
3	Network Connection	Win & Linux	Identifies socket bindings, destination IPs, ports, and initiating binary.
5	Process Terminated	Win & Linux	Monitors process lifecycle and malware termination behavior.
9	Raw Access Read	Linux	Detects direct device access (<code>/dev/sda</code> , memory scraping, credential dumps).
11	File Create	Win & Linux	Detects dropped malware payloads in <code>/tmp/</code> , <code>C:\Windows\Temp</code> , startup folders.
12/13	Registry Events	Windows	Monitors Run/RunOnce persistence keys and security policy tampering.
23	File Delete	Win & Linux	Tracks anti-forensic log wiping, ransomware file replacements.

5. Offensive Attack Simulations & Threat Scenarios

To rigorously validate the detection pipeline, multiple offensive security exercises were launched against the lab nodes using Kali Linux:

Scenario #	Attack Technique	Tool / Command	Target Node	Observed Wazuh Detection
AS-01	Host Discovery & TCP Port Scan	<code>nmap -sS -sV -O 10.0.2.20</code>	Windows 11	Nmap SYN scan alert & multi-port connection spike
AS-02	RDP Authentication Brute Force	<code>hydra -l admin -P wordlist rdp://</code>	Windows 11	Windows Event 4625 (Rule 18152: Logon Failure Threshold)
AS-03	Suspicious PowerShell CLI Execution	<code>powershell -enc ... -ExecutionPolicy Bypass</code>	Windows 11	Sysmon Event 1 & Custom Rule for Encoded PowerShell
AS-04	Linux Sudo & Privilege Escalation	<code>sudo su</code> & unauthorized user switches	Kali / Linux	Rule 5502: PAM Login session closed / authentication alert
AS-05	C2 Network Beacon Simulation	<code>nc -zv 10.0.2.15 4444</code> / egress tests	Linux Host	Sysmon Event 3 & Process Terminate Event 5 ingestion

6. Detection Engineering & MITRE ATT&CK; Mapping

Wazuh utilizes an XML-based ruleset architecture featuring over 3,000 built-in rules organized into severity levels ranging from Level 0 (Ignored) to Level 15 (Critical Alert).

Custom Detection Rules Implemented (`local\_rules.xml`)

To augment built-in coverage, specialized local rules were authored to detect advanced adversarial tactics:



MITRE ATT&CK; Framework Alignment

- **Reconnaissance (T1595):** Active scanning and port discovery detected via network connection frequency.
- **Initial Access / Credential Access (T1110):** Password brute-force detected via Event 4625 rate analysis.
- **Execution (T1059.001):** Command and Scripting Interpreter (PowerShell, bash, dash) captured by Sysmon Event ID 1.
- **Persistence (T1547):** Boot or Logon Autostart Execution monitored via Windows Registry key monitoring.
- **Defense Evasion (T1070):** Indicator Removal / File Deletion captured via Sysmon Event ID 23.

7. Ingestion Metrics, Threat Hunting & Analytical Results

During the evaluation window, the SOC Home Lab processed intensive event streams across Windows and Linux nodes, confirming high-throughput reliability without ingestion drops:

Telemetry Source	Agent Node	Total Events Ingested	Key Ingested Types	Alert Severity Levels
Sysmon for Linux (eBPF)	Kali Linux (004)	10,135+ Hits	Event ID 5, Event ID 1, Event ID 3	Level 3 to Level 10
Windows Sysmon & Events	Windows 11 (003)	1,427+ Hits	Event ID 1, 3, 4625, Registry, FIM	Level 3 to Level 12
Linux System & PAM Logs	Ubuntu / Kali	1,044+ Hits	Rule 5502, sudo sessions, sshd logs	Level 3 to Level 7
Total Lab Ingestion	All Multi-Nodes	12,600+ Events	Cross-platform telemetry correlation	Unified SIEM Dashboard

8. Operational Troubleshooting & Hardening

Real-world SOC deployments inevitably encounter communication roadblocks, agent disconnections, and parser bottlenecks. Key resolutions discovered during implementation include:

- **Agent Connectivity (Port 1514/1515):** Ensured Ubuntu UFW firewall explicitly permitted TCP traffic for ports 1514 and 1515. Resolved NAT gateway routing issues using consistent VirtualBox DHCP leases.
- **Sysmon Event Channel Ingestion:** Verified Windows Agent subscription to ``Microsoft-Windows-Sysmon/Operational`` using ``eventchannel`` log format in ``ossec.conf``.
- **Sysmon for Linux Kernel Dependencies:** Ensured Linux hosts satisfied minimum eBPF kernel requirements (≥ 5.4) and verified BPF virtual filesystem mounting via ``mount | grep bpf``.
- **Service Resiliency & Health Checks:** Established standardized PowerShell and bash recovery scripts (``NET START WazuhSvc``, ``systemctl restart wazuh-manager``) to ensure rapid service recovery.

9. Conclusion & Future Roadmap

This project successfully demonstrated the design, deployment, and practical operation of a comprehensive Wazuh-based SOC Home Lab. By unifying multi-platform endpoint telemetry (Windows Sysmon and Linux eBPF) into a centralized SIEM, the lab provides end-to-end visibility into host-level process creation, network activity, credential attacks, and unauthorized system modifications.

Future SOC Enhancements Roadmap

- **Network Intrusion Detection System (NIDS):** Integrate Suricata to capture full network packet payload inspection and signature-based exploit detection.
- **Threat Intelligence & Enrichment:** Connect VirusTotal, AbuseIPDB, and AlienVault OTX APIs into Wazuh integration pipelines for automated IoC scoring.
- **Security Orchestration, Automation, and Response (SOAR):** Deploy Shuffle SOAR to automate active response actions (e.g. automated host isolation upon high-severity alert triggers).
- **Incident Management:** Integrate TheHive and MISP for structured incident ticketing and threat intelligence sharing.

Report Authorization & Acknowledgement:

This report certifies the successful execution, empirical testing, and academic fulfillment of the SOC Home Lab capstone project.

Author: Natto Muni Chakma | **Institution:** Andhra University College of Engineering | **Status:** Approved & Completed